Explainable Machine Learning Framework for Chronic Kidney Disease Prediction: A SHAP-Based Interpretability Framework

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ABSTRACT

Artificial Intelligence (AI) and Machine Learning (ML) are increasingly being adopted in healthcare systems for disease prediction, clinical risk assessment, and decision-support applications. Although machine learning models can achieve strong predictive performance, many healthcare AI systems continue to operate as black-box models that provide limited insight into how predictions are generated. In clinical environments, predictive accuracy alone is insufficient, as healthcare professionals require transparent and interpretable explanations to support trust and informed decision-making.

Chronic Kidney Disease (CKD) is a major global health concern associated with progressive kidney function decline, increased healthcare burden, and elevated mortality risk. Early identification of CKD is essential for timely intervention and improved patient outcomes. Existing CKD prediction studies have largely focused on improving predictive performance, while comparatively less attention has been given to model interpretability and clinical transparency.

This study presents an Explainable Artificial Intelligence (XAI) framework for Chronic Kidney Disease prediction using a Decision Tree classifier integrated with SHapley Additive Explanations (SHAP) [1]. The research utilizes the Chronic Kidney Disease dataset from the UCI Machine Learning Repository, consisting of 400 patient records and multiple clinical attributes related to kidney function assessment. Data preprocessing techniques, including missing value imputation, categorical encoding, and numerical transformation, were applied prior to model development.

The proposed framework combines predictive modelling with explainability analysis to provide both accurate disease prediction and transparent interpretation of model behaviour. Experimental results demonstrated strong predictive performance, achieving high accuracy, ROC-AUC score, and cross-validation performance. SHAP analysis identified clinically significant features, including Hemoglobin, Specific Gravity, Blood Urea, Serum Creatinine, and Hypertension, as major contributors influencing CKD prediction outcomes.

The findings demonstrate that integrating machine learning with explainability techniques can improve transparency, trustworthiness, and interpretability in healthcare AI systems. The proposed framework contributes to the development of more reliable and clinically understandable decision-support tools for early CKD prediction and healthcare analytics.

Keywords — *Explainable AI, Machine Learning, SHAP, Chronic Kidney Disease, Healthcare Analytics, Clinical Prediction, Decision Tree, Interpretable Machine Learning*

1. INTRODUCTION

Artificial Intelligence and Machine Learning are increasingly being adopted in healthcare systems for disease prediction, clinical risk assessment, and medical decision-support systems [3],[4]. Predictive healthcare models can analyze patient health records and identify patterns associated with disease progression. Chronic Kidney Disease (CKD) has become a major global healthcare concern due to its increasing prevalence and long-term health complications.

Early detection of CKD is critical because delayed diagnosis may lead to kidney failure, cardiovascular complications, and increased mortality risk. Machine learning approaches can assist healthcare professionals by enabling early-stage prediction using clinical biomarkers and patient health information.

Despite the growing use of machine learning in healthcare, many predictive models operate as black-box systems where the internal reasoning behind predictions remains unclear. In healthcare environments, predictive accuracy alone is insufficient because clinicians must understand why a model generates a particular prediction before integrating it into medical practice.

Explainable Artificial Intelligence (XAI) techniques address this challenge by improving transparency and interpretability in AI systems [8],[10]. SHAP (SHapley Additive Explanations) is one of the most widely used explainability frameworks because it provides both local and global explanations regarding feature contribution and model behavior[1].

This research focuses on developing an explainable machine learning framework for Chronic Kidney Disease prediction using Decision Tree classification integrated with SHAP interpretability analysis.

1.1 Research Gap

Machine learning techniques have demonstrated promising results in Chronic Kidney Disease prediction and clinical risk assessment. However, many existing studies primarily emphasize predictive performance while providing limited insight into the reasoning behind model decisions. This lack of interpretability creates challenges for clinical adoption, as healthcare professionals require transparent explanations before integrating AI-assisted predictions into medical decision-making processes. Furthermore, many predictive models operate as black-box systems that do not clearly identify which clinical features contribute most significantly to disease outcomes. Therefore, there remains a need for explainable machine learning frameworks that combine accurate disease prediction with transparent and clinically meaningful interpretation.

1.2 Contributions of this Study

The main contributions of this research are as follows:

- Development of an explainable machine learning framework for Chronic Kidney Disease prediction.
- Implementation of a Decision Tree classification model using clinical healthcare data.
- Integration of SHAP-based explainability analysis to provide both global and local model interpretation.
- Identification of clinically important biomarkers influencing CKD prediction outcomes.
- Evaluation of predictive performance using multiple classification metrics and cross-validation.
- Demonstration of how Explainable AI can improve transparency, trust, and interpretability in healthcare decision-support systems.

2. OBJECTIVES

- To analyze the Chronic Kidney Disease healthcare dataset.
- To preprocess and clean clinical healthcare data.
- To implement a Decision Tree classifier for CKD prediction.
- To evaluate model performance using standard classification metrics.

- To apply SHAP explainability techniques for model interpretation.
- To identify clinically important features contributing to CKD prediction.
- To improve transparency and trust in healthcare AI systems.

3. RELATED WORK

Machine learning techniques have been widely adopted for healthcare prediction tasks, particularly for the early detection and diagnosis of chronic diseases [3],[4]. Chronic Kidney Disease prediction has attracted significant research attention due to its increasing prevalence and the importance of early intervention. Several studies have applied machine learning algorithms such as Logistic Regression, Decision Trees, Random Forests, Support Vector Machines, Naïve Bayes, and Artificial Neural Networks to improve CKD prediction performance [11].

Previous research has demonstrated that machine learning models can effectively identify patterns within clinical and laboratory data for CKD diagnosis. Ensemble-based approaches such as Random Forest and Gradient Boosting have often achieved strong predictive performance because of their ability to capture complex relationships among healthcare variables. However, many of these models operate as black-box systems, making it difficult for healthcare professionals to understand the reasoning behind predictions [8],[9].

In recent years, Explainable Artificial Intelligence (XAI) has emerged as an important research area aimed at improving transparency and trust in machine learning systems. Among various explainability techniques, SHAP (SHapley Additive Explanations) has gained significant attention due to its strong theoretical foundation and ability to provide both local and global explanations. SHAP assigns contribution values to individual features, enabling researchers and clinicians to understand how specific variables influence prediction outcomes.

Several healthcare studies have successfully applied SHAP to improve interpretability in disease prediction models, including diabetes prediction, cardiovascular disease risk assessment, cancer diagnosis, and kidney disease analysis. These studies demonstrated that explainability methods can help identify clinically relevant biomarkers while increasing confidence in AI-assisted decision-support systems.

Despite these advancements, relatively few studies have focused on combining interpretable machine learning models with SHAP-based explainability specifically for Chronic Kidney Disease prediction. This study addresses this gap by integrating a Decision Tree classifier with SHAP explainability analysis to provide both accurate prediction performance and clinically interpretable insights.

4. DATASET DESCRIPTION

The Chronic Kidney Disease dataset used in this study was obtained from the UCI Machine Learning Repository [2],[7]. The dataset contains clinical and laboratory records collected from patients for CKD diagnosis analysis.

4.1 Dataset Overview

- Total Instances: 400 patients
- Total Features: 25 clinical attributes
- Task Type: Binary Classification
- Target Variable: CKD / Not CKD

4.2 Important Clinical Features

- Hemoglobin (hemo)
- Specific Gravity (sg)
- Blood Urea (bu)
- Serum Creatinine (sc)

- Packed Cell Volume (pcv)
- Hypertension (htn)
- Red Blood Cell Count (rc)
- Potassium (pot)

Target Variable

- ckd → Chronic Kidney Disease
- notckd → No Chronic Kidney Disease

4.3 Dataset Characteristics

The dataset contains both numerical and categorical clinical attributes commonly used during kidney disease diagnosis and monitoring. Several attributes contain missing values, reflecting real-world healthcare data collection challenges. Therefore, appropriate preprocessing techniques were required before model development.

The dataset is suitable for supervised binary classification tasks where the objective is to predict whether a patient is likely to have Chronic Kidney Disease based on available clinical measurements and laboratory test results. The diversity of clinical variables included in the dataset provides an opportunity to investigate both predictive performance and feature-level interpretability using Explainable AI techniques.

4.4 Class Distribution

The dataset contains both CKD and non-CKD patient records, providing a balanced benchmark for binary classification tasks. The presence of heterogeneous clinical measurements enables comprehensive evaluation of machine learning models for healthcare prediction while also supporting interpretability analysis through explainable AI techniques.

4.5 Dataset Limitations

Several important limitations of this dataset must be acknowledged. First, the dataset contains only 400 records, which is relatively small for machine learning tasks and may lead to overfitting, particularly for models with high representational capacity. Second, all data originates from a single hospital in southern India, limiting geographic and demographic generalizability. Third, missing values are present across several features, requiring imputation assumptions that may introduce bias. Fourth, the dataset was collected in 2015 and may not reflect current clinical practices or patient demographics. These limitations are reflected in cautious interpretation of model performance throughout this paper.

5. METHODOLOGY

5.1 Data Collection

The CKD dataset was imported into the Python analysis environment using Pandas [3].

5.2 Data Preprocessing

The preprocessing stage included:

- Removal of unnecessary identifier columns
- Conversion of categorical variables into numerical form
- Missing value handling using SimpleImputer
- Feature transformation
- Train-test splitting for model training and evaluation

5.3 Machine Learning Model

A Decision Tree classifier was implemented using Scikit-learn [3].

The model was trained using:

- Training Dataset: 80%
- Testing Dataset: 20%

5.4 Explainability Analysis

SHAP explainability analysis was performed to understand feature-level contributions influencing CKD prediction outcomes [1].

SHAP analysis enabled:

- Global model interpretation [1]
- Local prediction explanation [1]
- Feature importance analysis [1]
- Transparent healthcare AI interpretation [11]

5.5 Model Evaluation Strategy

To assess the effectiveness of the proposed framework, multiple evaluation metrics were employed. Accuracy was used to measure the overall prediction performance of the model, while Precision and Recall were used to evaluate the model's ability to correctly identify CKD cases and minimize misclassification.

The F1-Score was calculated to provide a balanced assessment of Precision and Recall performance. Additionally, Receiver Operating Characteristic (ROC) analysis and Area Under the Curve (ROC-AUC) were used to evaluate the model's discriminative capability across different classification thresholds.

To further assess model stability and generalization performance, k-fold cross-validation was performed. Cross-validation reduces the risk of performance estimation bias and provides a more reliable assessment of model robustness.

The overall workflow of the proposed framework consists of data collection, preprocessing, model development, performance evaluation, and SHAP-based explainability analysis.

6. MACHINE LEARNING IMPLEMENTATION

The Decision Tree classifier was trained using the pre-processed CKD dataset. The model successfully learned relationships between clinical variables and disease outcomes.

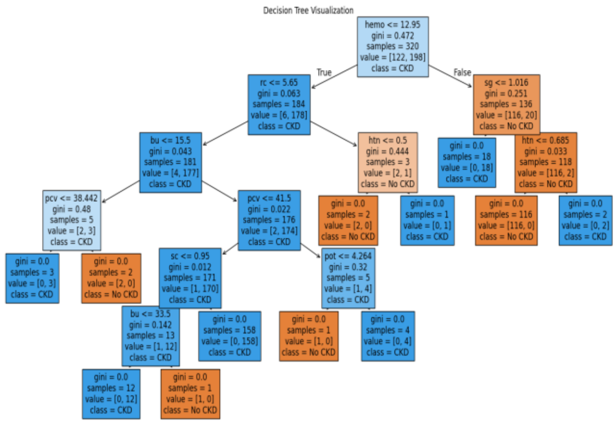

Fig 1: Decision Tree structure used for CKD prediction

6.1 Model Selection Rationale

The Decision Tree classifier was selected because of its inherent interpretability and suitability for healthcare applications where transparency is essential. Unlike many black-box machine learning models, Decision Trees provide human-understandable decision paths that can be interpreted by clinicians and healthcare professionals [8],[9].

Additionally, the model works effectively with mixed healthcare data containing both numerical and categorical variables. The integration of Decision Tree classification with SHAP explainability further enhances transparency by providing feature-level explanations for prediction outcomes.

7. RESULTS AND PERFORMANCE ANALYSIS

The model was evaluated using multiple classification metrics.

Performance Metrics:

- Accuracy
- Precision
- Recall
- F1 Score
- ROC-AUC Score

- Cross Validation

EvaluationMetric	Value
Accuracy	1.00
Precision	1.00
Recall	1.00
F1-Score	1.00
ROC-AUC Score	1.00
Average Cross Validation Score	0.97

7.1 Performance Interpretation

The proposed Decision Tree model achieved excellent predictive performance across all evaluation metrics. The model successfully distinguished between CKD and non-CKD patients with very high classification accuracy. Although near-perfect performance was observed, the results should be interpreted cautiously because the dataset size is relatively small and may not fully represent real-world clinical variability. The observed performance may also reflect the relatively clean structure of the dataset and should therefore be validated on larger external clinical datasets before generalizing the findings.

The ROC-AUC score indicates strong discriminative capability, demonstrating that the model can effectively separate positive and negative classes across different classification thresholds. Furthermore, the high cross-validation score suggests that the model maintains stable performance across multiple training and testing splits, indicating good generalization ability.

These results suggest that machine learning can effectively support early-stage CKD prediction using routinely collected clinical and laboratory measurements.

7.2 Limitations of Performance Evaluation

Although the model achieved excellent performance on the available dataset, the results should be interpreted with caution. The dataset contains only 400 patient records collected from a single source, which may limit the generalizability of the findings. In addition, the near-perfect evaluation metrics may partially reflect characteristics of the dataset and its relatively limited complexity. Therefore, external validation using larger, more diverse, and multi-center clinical datasets is necessary before drawing conclusions regarding real-world clinical deployment and robustness.

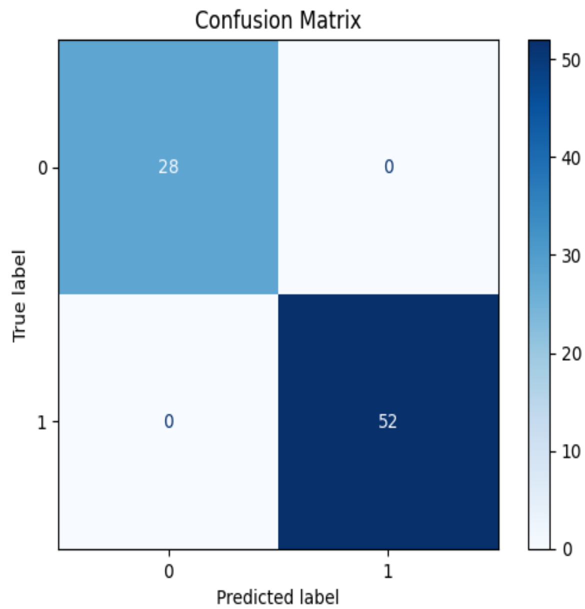

Fig 2: Confusion Matrix for Decision Tree classification model

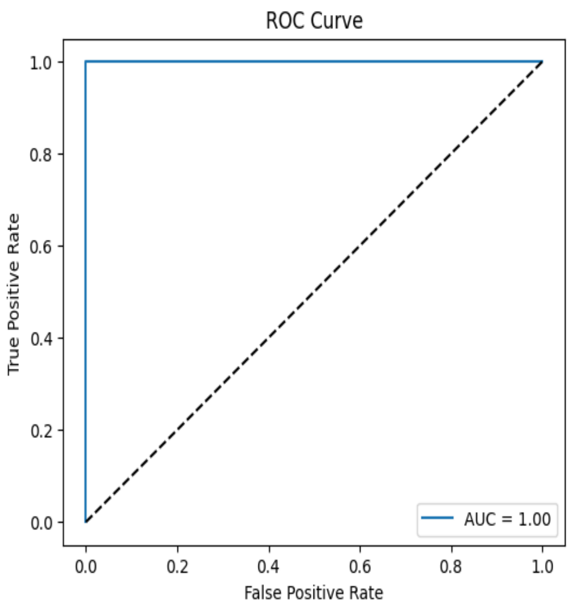

Fig 3: ROC Curve showing classification performance of the Decision Tree model

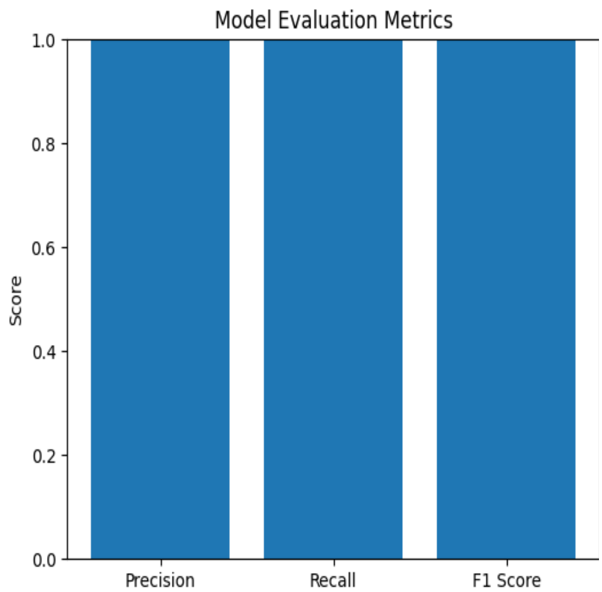

Fig 4: Evaluation metrics comparison for the proposed model

Observation:

The results demonstrate strong predictive performance for CKD classification.

8. SHAP EXPLAINABILITY ANALYSIS

SHAP analysis was performed to identify clinically significant features influencing CKD prediction outcomes [1].

Important Features Identified:

- Hemoglobin (hemo)
- Specific Gravity (sg)
- Hypertension (htn)
- Blood Urea (bu)
- Packed Cell Volume (pcv)

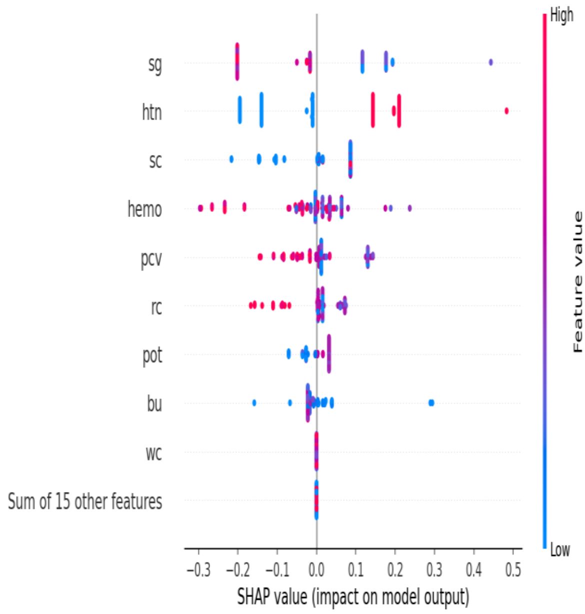

Fig 5: SHAP Summary Plot showing feature contribution to CKD prediction

8.1 Clinical Interpretation of SHAP Results

The SHAP analysis revealed that Hemoglobin, Specific Gravity, Blood Urea, Serum Creatinine, and Hypertension were among the most influential variables affecting model predictions.

Lower Hemoglobin levels and abnormal Specific Gravity values were frequently associated with increased CKD prediction probability. Similarly, elevated Blood Urea and Serum Creatinine levels contributed significantly toward positive CKD classifications, which is consistent with established clinical understanding of kidney dysfunction.

The SHAP framework not only identified important predictive features but also quantified their individual contributions to model decisions, improving transparency and supporting clinical interpretability.

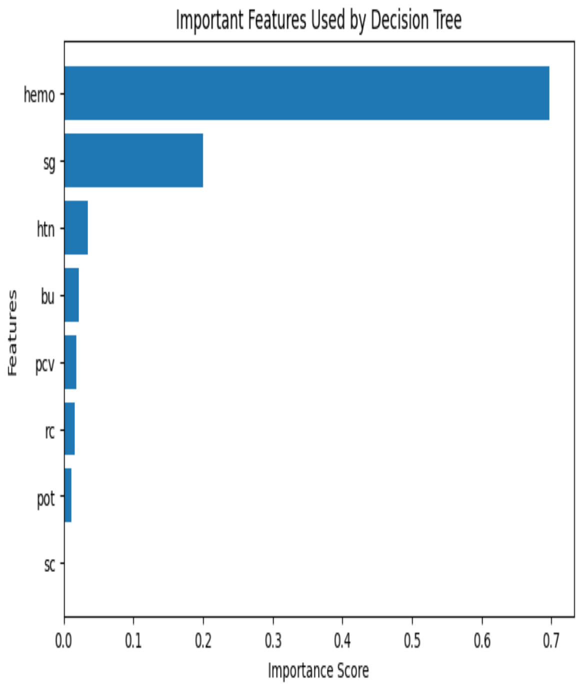

Fig 6: Global feature importance using SHAP values

The SHAP explainability framework improved transparency and interpretability by identifying how feature values positively or negatively influenced model predictions.

9. DISCUSSION

The results obtained in this study demonstrate the potential of combining machine learning and explainable artificial intelligence techniques for healthcare prediction tasks [11],[13]. While the Decision Tree model achieved strong predictive performance, the primary contribution of this work lies in improving model interpretability through SHAP-based explanations.

Healthcare applications require transparency because clinical decisions directly affect patient outcomes. Black-box machine learning models may achieve high predictive accuracy but often face resistance in real-world healthcare adoption due to limited interpretability. By integrating SHAP explainability, the proposed framework provides insights into how individual clinical variables influence prediction outcomes [1],[10].

The identified important features align with established medical knowledge regarding Chronic Kidney Disease progression, suggesting that the model captures clinically meaningful relationships within the dataset. This increases confidence in the reliability and trustworthiness of the developed framework.

Although the model achieved near-perfect predictive performance on the available dataset, the relatively small dataset size may contribute to optimistic evaluation results. Future work should evaluate the proposed framework using larger and externally validated clinical datasets to assess generalizability in real-world healthcare settings [11],[13].

Overall, the study demonstrates that explainability techniques can bridge the gap between predictive performance and clinical interpretability, supporting more transparent healthcare decision-making processes. The combination of predictive performance and interpretability is particularly important in healthcare settings where clinical decisions require both accuracy and explainability.

10. CONCLUSION

This research proposed an explainable machine learning framework for Chronic Kidney Disease prediction using a Decision Tree classifier integrated with SHAP-based interpretability analysis.

The model achieved strong predictive performance while maintaining transparency and interpretability through SHAP explainability techniques. Important healthcare features including Hemoglobin, Specific Gravity, Blood Urea, and Hypertension were identified as significant contributors influencing CKD prediction outcomes.

The study demonstrates that Explainable AI can improve trustworthiness and transparency in healthcare prediction systems, supporting more reliable clinical decision-making processes [11],[13].

The integration of machine learning and explainable AI techniques provides a promising direction for developing trustworthy healthcare decision-support systems. By combining predictive performance with transparent explanations, healthcare professionals can better understand model behavior and make more informed clinical decisions. Future development of explainable clinical prediction systems may contribute to safer and more transparent AI-assisted healthcare environments.

11. FUTURE WORK

Future research can extend this work by evaluating additional machine learning algorithms such as Random Forest, XGBoost, LightGBM, and deep learning models. Comparative analysis across multiple explainability techniques, including SHAP, LIME, and Integrated Gradients, may provide further insights into model interpretability [14].

The framework can also be validated using larger multi-center healthcare datasets to improve generalizability and robustness. Future studies may explore real-time deployment within clinical decision-support systems and investigate the integration of longitudinal patient records for improved disease progression monitoring.

Furthermore, the proposed explainable AI framework can be extended to other chronic disease prediction tasks, including diabetes, cardiovascular disease, and liver disease risk assessment.

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